Zero-Mode-Safe Orthogonal Coarsening:
Exact Projection Monotonicity, Constant-Defect Sharpness, and the Critical J^{-1} Interface

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Abstract

Let Ω be a prescribed finite support of size S, let $A \in \mathbb{C}^{\Omega}$, and put

$$Z(A) = \langle A, \mathbf{1} \rangle, \qquad D(A) = \|A\|_2^2, \qquad \rho(A) = \frac{|Z(A)|}{\sqrt{SD(A)}}, \qquad \mathfrak{F}(A) = \frac{|Z(A)|^2}{D(A)}.$$

For an orthogonal projection P, write

$$B = PA, \qquad R = (I-P)A, \qquad q = (I-P)\mathbf{1}.$$

The exact projection ledger is

$$D(A) = D(B) + D(R), \qquad Z(B) - Z(A) = -\langle R, q \rangle = \langle A, (P-I)\mathbf{1} \rangle.$$

If $P\mathbf{1} = \mathbf{1}$, then $q = 0$, so the zero mode is preserved exactly while energy can only decrease. Whenever $B \neq 0$,

$$\rho(A)^2 = \rho(B)^2 \frac{D(B)}{D(A)}, \qquad \mathfrak{F}(A) = \mathfrak{F}(B) \frac{D(B)}{D(A)}.$$

Thus

$$\rho(A) \le \rho(B), \qquad \mathfrak{F}(A) \le \mathfrak{F}(B).$$

Constant-preserving orthogonal coarsening is therefore a *conservative* flatness surrogate: it may create false negatives, but cannot create a false positive by artificially reducing the physical zero-mode ratio. This gives a structural exception to the general J^{-1} surrogate-accuracy barrier: a one-sided flatness certificate transfers without requiring $\|A - PA\|_2 / \|PA\|_2 = O(J^{-1})$.

For a partition of the actual support, ordinary block averaging is exactly such a projection. We prove the within-block variance identity, monotonicity along nested partitions, and the precise weighted compressed representation. Replacing block multiplicities by an unweighted compressed grid can manufacture perfect apparent cancellation; a three-point counterexample shows that the actual-support convention is essential.

If $P\mathbf{1} \neq \mathbf{1}$, define the normalized constant defect

$$\kappa(P) = \frac{\|(I-P)\mathbf{1}\|_2}{\sqrt{S}}.$$

Then the sharp structure-free estimate is

$$|Z(B) - Z(A)| \le \sqrt{S}\,\kappa(P)\|R\|_2.$$

For $B \neq 0$, with $r = \|R\|_2 / \|B\|_2$,

$$\frac{(\rho(B) - \kappa(P)r)_+}{\sqrt{1+r^2}} \le \rho(A) \le \frac{\rho(B) + \kappa(P)r}{\sqrt{1+r^2}}.$$

Equality in the zero-mode bound is attained by taking the residual parallel to the missing constant direction. Consequently, retaining a critical J^{-1} zero-mode scale from norm information alone requires

$$\kappa(P)r = O(J^{-1});$$

when $\kappa(P)$ stays bounded below, the TPC-89 $O(J^{-1})$ relative-accuracy barrier returns. We also give the unique minimal rank-one completion

$$P^\sharp = P + \frac{\langle\,\cdot\,,q\rangle}{\|q\|_2^2}q$$

that adjoins the missing constant mode.

On the literal fixed-h_0 TPC determinant frame, any constant-preserving orthogonal surrogate satisfying the projected critical bound immediately certifies the same bound for the physical coefficient. This is an exact L1 certificate. The paper constructs no arithmetic projection estimate and proves no new L2 fixed-shift cancellation theorem.

Keywords: zero mode; orthogonal projection; block averaging; conditional expectation; determinant flatness; actual support; surrogate; constant-mode defect.

Literal-frame convention. Every physical statement is made on the actual post-bin support of the already aggregated coefficient for one prescribed nonzero shift h_0, with its native bins and one global atomic normalization. A block-compressed vector is interpreted with its exact block multiplicities. No statement is specialized to $h_0 = 2$.

1 Introduction

The determinant-flatness route separates one signed coefficient into three literal statistics:

$$\text{zero mode } Z, \qquad \text{energy } D, \qquad \text{actual support size } S.$$

TPC-88 gives the exact mass–cancellation factorization, while TPC-89 identifies

$$\rho = \frac{|Z|}{\sqrt{SD}}$$

as the projective angle to the constant line [3, 4]. For a general same-frame surrogate B, TPC-89 proves a sharp worst-case fact: transferring the critical target $\rho \ll J^{-1}$ from only a relative ℓ^2 approximation requires

$$\frac{\|A-B\|_2}{\|B\|_2} = O(J^{-1}).$$

An $o(1)$ approximation need not suffice.

That theorem treats the error as an arbitrary direction. The present paper asks whether a structured surrogate can avoid this worst case. The answer is exact.

Suppose $B = PA$, where P is an orthogonal projection on the *same prescribed physical frame*. If the constant vector lies in $\operatorname{Ran} P$, then the residual is automatically orthogonal to the constant vector. Hence

$$Z(PA) = Z(A)$$

with no approximation, while Pythagoras gives

$$D(PA) \le D(A).$$

The projected flatness is therefore at least the physical flatness. This reverses the usual concern about surrogate optimism: constant-preserving orthogonal coarsening is pessimistic in exactly the direction needed for a one-sided upper certificate.

The observation is elementary, but three distinctions make its literal use nontrivial.

(i) The projection must act on the actual post-bin frame. A compressed block vector must retain block multiplicities.

(ii) Preserving constants as an output, $P\mathbf{1} = \mathbf{1}$, is enough here only because $P = P^*$. For a general linear surrogate the zero-mode condition is $T^*\mathbf{1} = \mathbf{1}$.

(iii) A flat projected coefficient is a valid certificate only after its own arithmetic bound has been proved. Projection monotonicity does not estimate that coefficient.

If the projection misses the constant vector, the exact obstruction is not the entire residual but one scalar pairing:

$$Z(PA) - Z(A) = \langle A, (P - I)\mathbf{1}\rangle.$$

Writing $q = (I - P)\mathbf{1}$, orthogonality sharpens this to

$$Z(PA) - Z(A) = -\langle (I - P)A, q\rangle.$$

The resulting bound is governed by the product of two dimensionless quantities:

$$\kappa(P) = \frac{\|q\|_2}{\sqrt{S}}, \qquad r(A; P) = \frac{\|(I - P)A\|_2}{\|PA\|_2}.$$

The critical interface becomes

$$\kappa(P)r(A; P) \ll J^{-1}.$$

This interpolates between the exact safe case $\kappa = 0$ and the general TPC-89 barrier when $\kappa \asymp 1$.

1.1 Main contributions

The paper proves the following finite-dimensional statements.

(1) An exact zero-mode/energy ledger for every orthogonal projection, including the defect identity and its sharp equality cases.

(2) Exact flatness monotonicity for every constant-preserving projection, with a multiplicative formula and equality classification.

(3) A monotone hierarchy for nested coarse-grainings and a within-block variance formula for partition averages.

(4) A weighted compression theorem and an explicit counterexample to unweighted block compression.

(5) Sharp constant-defect bounds and the product criterion $\kappa r = O(J^{-1})$.

(6) The minimal rank-one constant completion of an unsafe projection.

(7) An exact L1 crosswalk to the literal fixed-h_0 determinant coefficient and the TPC-92/TPC-93 continuation program [5, 6].

1.2 Logical level

The projection identities, sharpness constructions, partition laws, and rank-one completion are unconditional L0 results. Applying them to a fully specified operator on the literal TPC determinant bins is an L1 certificate interface. No operator is shown here to have a small projected zero-mode ratio for the growing fixed-h_0 arithmetic family. Consequently there is no new L2 arithmetic estimate, no parity breakthrough, and no prime-pair conclusion.

1.3 Organization

Section 2 fixes the frame and statistics. Section 3 proves the safe projection theorem. Section 4 treats literal block averages and compression. Section 5 gives the sharp defective case, and section 6 repairs it. Section 7 isolates the J^{-1} threshold. Section 8 gives the fixed-shift attachment, section 9 records the executable finite certificate, and section 10 states the exact claim boundary.

2 The prescribed finite frame

Let Ω be a nonempty finite set and put

$$S = \#\Omega, \qquad H = \mathbb{C}^{\Omega}.$$

Equip H with the counting-measure inner product

$$\langle x, y\rangle = \sum_{\omega\in\Omega} x_\omega \overline{y_\omega},$$

linear in the first variable. Let $\mathbf{1}$ be the constant vector, so $\|\mathbf{1}\|_2 = \sqrt{S}$, and put

$$e = \frac{\mathbf{1}}{\sqrt{S}}.$$

Definition 2.1 (Literal zero-mode statistics)**.** For $A \in H$, define

$$Z(A) = \langle A, \mathbf{1}\rangle = \sum_{\omega\in\Omega} A_\omega, \qquad D(A) = \|A\|_2^2.$$

For $A \neq 0$, define

$$\rho(A) = \frac{|Z(A)|}{\sqrt{S}\|A\|_2}, \qquad \mathfrak{F}(A) = \frac{|Z(A)|^2}{D(A)} = S\rho(A)^2.$$

The zero vector is a separate branch; no quotient by $D(0)$ is used.

Remark 2.2 (The frame is part of the statistic)*.* The quantity $\mathfrak{F}$ is unchanged by zero padding, but ρ is not. If K zero coordinates are appended, then

$$\rho_{\text{padded}}(A) = \sqrt{\frac{S}{S+K}}\,\rho_\Omega(A).$$

Thus every statement about ρ requires a prescribed frame. Here the frame is always the actual support or an exactly weighted representation of it, as in section 4.

Let $P : H \to H$ be an orthogonal projection:

$$P^2 = P, \qquad P^* = P.$$

For a fixed A, write

$$B = PA, \qquad R = (I - P)A. \tag{1}$$

For the constant vector, write

$$p = P\mathbf{1}, \qquad q = (I - P)\mathbf{1}, \qquad \kappa(P) = \frac{\|q\|_2}{\sqrt{S}}. \tag{2}$$

Lemma 2.3 (Orthogonal ledger)**.** *The decompositions*

$$A = B + R, \qquad \mathbf{1} = p + q$$

are orthogonal. In particular,

$$D(A) = D(B) + D(R), \tag{3}$$
$$S = \|p\|_2^2 + \|q\|_2^2, \tag{4}$$
$$0 \le \kappa(P) \le 1. \tag{5}$$

Moreover,

$$B, p \in \operatorname{Ran} P, \qquad R, q \in \operatorname{Ker} P = (\operatorname{Ran} P)^\perp.$$

Proof. These are the standard range–kernel decompositions for an orthogonal projection [1, Chapter I]. Applying Pythagoras first to A and then to $\mathbf{1}$ gives (3) and (4). $\square$

Lemma 2.4 (Zero-mode defect identity)**.** *For every $A \in H$,*

$$\boxed{Z(B) - Z(A) = \langle A, (P - I)\mathbf{1}\rangle = -\langle R, q\rangle.} \tag{6}$$

Equivalently,

$$Z(A) = \langle B, p\rangle + \langle R, q\rangle, \qquad Z(B) = \langle B, p\rangle. \tag{7}$$

Proof. Self-adjointness gives

$$Z(B) = \langle PA, \mathbf{1}\rangle = \langle A, P\mathbf{1}\rangle.$$

Subtracting $\langle A, \mathbf{1}\rangle$ gives the first identity in (6). Since $(P - I)\mathbf{1} = -q$ and $B \perp q$,

$$\langle A, q\rangle = \langle B + R, q\rangle = \langle R, q\rangle.$$

The split formulas follow by expanding $\langle B + R, p + q\rangle$. $\square$

Proposition 2.5 (The exact preservation criterion)**.** *The following conditions are equivalent:*

(i) $P\mathbf{1} = \mathbf{1}$*;*

(ii) $\mathbf{1} \in \operatorname{Ran} P$*;*

(iii) $\operatorname{Ker} P \subset \mathbf{1}^{\perp}$*;*

(iv) $Z(PA) = Z(A)$ *for every* $A \in H$*.*

Proof. The equivalence of (i) and (ii) follows because an orthogonal projection fixes precisely its range. Conditions (ii) and (iii) are equivalent because $\operatorname{Ker} P = (\operatorname{Ran} P)^{\perp}$. Condition (i) implies (iv) by lemma 2.4. Conversely, if (iv) holds, then

$$\langle A, (P - I)\mathbf{1}\rangle = 0$$

for every A, hence $(P - I)\mathbf{1} = 0$. $\square$

Remark 2.6 (General operators)*.* For a general linear map T, the correct condition is

$$T^*\mathbf{1} = \mathbf{1},$$

not merely $T\mathbf{1} = \mathbf{1}$. Orthogonal projections identify the two because $P^* = P$. The present monotonicity theorem also uses orthogonal energy splitting, so it should not be transferred to an arbitrary averaging matrix without separately checking adjoint mass preservation and norm contraction.

3 Constant-preserving projections are conservative

We now impose the exact safe condition

$$P\mathbf{1} = \mathbf{1}.$$

Then $q = 0$, so every discarded residual has zero sum.

Theorem 3.1 (Exact safe-projection law)**.** *Let P be an orthogonal projection with $P\mathbf{1} = \mathbf{1}$, and let $A \neq 0$. Put $B = PA$ and $R = (I - P)A$. Then*

$$\boxed{Z(B) = Z(A), \qquad D(A) = D(B) + D(R).} \tag{8}$$

If $B = 0$, then $Z(A) = 0$ and hence $\rho(A) = \mathfrak{F}(A) = 0$. If $B \neq 0$, then

$$\boxed{\rho(A)^2 = \rho(B)^2 \frac{D(B)}{D(A)}, \qquad \mathfrak{F}(A) = \mathfrak{F}(B)\frac{D(B)}{D(A)}.} \tag{9}$$

Consequently,

$$\boxed{\rho(A) \leq \rho(B), \qquad \mathfrak{F}(A) \leq \mathfrak{F}(B).} \tag{10}$$

Proof. The first formula follows from proposition 2.5; the second is (3). If $B = 0$, zero-mode preservation gives $Z(A) = Z(B) = 0$. If $B \neq 0$, insert the common numerator $|Z(A)|^2 = |Z(B)|^2$ into the definitions:

$$\frac{\rho(A)^2}{\rho(B)^2} = \frac{D(B)}{D(A)}, \qquad \frac{\mathfrak{F}(A)}{\mathfrak{F}(B)} = \frac{D(B)}{D(A)}.$$

The conclusion follows from $D(B) \leq D(A)$. When the common zero mode vanishes, both sides of the identities are zero. □

Corollary 3.2 (Exact residual factor)**.** *Under the hypotheses of theorem 3.1, suppose $B \neq 0$, and put*

$$r = \frac{\|R\|_2}{\|B\|_2}.$$

Then

$$\rho(A) = \frac{\rho(B)}{\sqrt{1+r^2}}, \qquad \mathfrak{F}(A) = \frac{\mathfrak{F}(B)}{1+r^2}. \tag{11}$$

Proof. By Pythagoras,

$$\frac{D(A)}{D(B)} = 1 + r^2.$$

Apply (9). □

Corollary 3.3 (One-sided certificate without approximation)**.** *Let $T \geq 0$. Under the hypotheses of theorem 3.1,*

$$\mathfrak{F}(B) \leq T \quad \Longrightarrow \quad \mathfrak{F}(A) \leq T,$$

where the branch $B = 0$ is interpreted by the conclusion $\mathfrak{F}(A) = 0$. Likewise, on the same frame,

$$\rho(B) \leq \tau \quad \Longrightarrow \quad \rho(A) \leq \tau.$$

No upper bound on $\|A - B\|_2/\|B\|_2$ is required.

Proof. This is (10), including the zero branch in theorem 3.1. □

Remark 3.4 (Why this does not contradict TPC-89)*.* TPC-89 controls an arbitrary same-frame error. Here

$$A - B = R \perp \operatorname{Ran} P, \qquad \mathbf{1} \in \operatorname{Ran} P,$$

so $R \perp \mathbf{1}$. The dangerous tangent direction toward the constant line is excluded exactly, not approximately. This extra orthogonality is the structural information that TPC-89 explicitly allows as an escape from its worst-case J^{-1} barrier [4].

Proposition 3.5 (Equality classification)**.** *Suppose $P\mathbf{1} = \mathbf{1}$, $A \neq 0$, and $B = PA \neq 0$.*

(i) If $Z(A) = 0$, then $\rho(A) = \rho(B) = \mathfrak{F}(A) = \mathfrak{F}(B) = 0$, regardless of R.

(ii) If $Z(A) \neq 0$, equality in either inequality of (10) holds if and only if $R = 0$, equivalently $A \in \operatorname{Ran} P$.

Proof. The first statement follows from zero-mode preservation. In the second branch, (9) has nonzero numerator. Equality is equivalent to $D(A) = D(B)$, and by Pythagoras this is equivalent to $D(R) = 0$. □

Proposition 3.6 (Safe energy and zero-mode dominance)**.** *Suppose $P\mathbf{1} = \mathbf{1}$. Any bounds proved for $B = PA$ of the form*

$$D(B) \geq L, \qquad |Z(B)| \leq U$$

imply, for the original coefficient,

$$D(A) \geq L, \qquad |Z(A)| \leq U.$$

Thus a constant-preserving projected packet may simultaneously provide a determinant-energy lower certificate and a zero-mode upper certificate.

Proof. Use $D(A) \geq D(B)$ and $Z(A) = Z(B)$. $\square$

Remark 3.7 (Conservative does not mean easy). Coarsening reduces the denominator while preserving the numerator, so it usually makes ρ and $\mathfrak{F}$ larger. A projected upper bound is therefore safe but potentially harder to prove. Projection monotonicity supplies no cancellation estimate for B; it only guarantees that a proved estimate cannot be a false positive created by the coarsening.

4 Block averaging on the actual support

Let

$$\mathcal{B} = \{B_1, \dots, B_m\}$$

be a partition of Ω into nonempty blocks. For $\omega \in B_j$, define

$$(P_{\mathcal{B}}A)_\omega = \overline{A}_j := \frac{1}{|B_j|} \sum_{\nu \in B_j} A_\nu. \tag{12}$$

Proposition 4.1 (Block averaging is a safe projection)**.** *The operator $P_{\mathcal{B}}$ is the orthogonal projection onto the subspace of vectors that are constant on each block. Moreover,*

$$P_{\mathcal{B}}\mathbf{1} = \mathbf{1}.$$

Consequently,

$$Z(P_{\mathcal{B}}A) = Z(A), \tag{13}$$

$$D(P_{\mathcal{B}}A) = \sum_{j=1}^{m} |B_j|\, |\overline{A}_j|^2, \tag{14}$$

$$D(A) - D(P_{\mathcal{B}}A) = \sum_{j=1}^{m} \sum_{\omega \in B_j} |A_\omega - \overline{A}_j|^2. \tag{15}$$

If $P_{\mathcal{B}}A \neq 0$, then

$$\mathfrak{F}(A) \leq \mathfrak{F}(P_{\mathcal{B}}A), \qquad \rho(A) \leq \rho(P_{\mathcal{B}}A),$$

where both statistics use the original frame size S.

Proof. The formula (12) is idempotent and its range is the block-constant subspace. For every block-constant C,

$$\langle A - P_{\mathcal{B}}A, C \rangle = \sum_{j=1}^{m} \overline{C_j} \sum_{\omega \in B_j} (A_\omega - \overline{A}_j) = 0.$$

Hence the projection is orthogonal. It clearly fixes $\mathbf{1}$. Equation (13) follows by summing block means with their multiplicities. Equations (14) and (15) are Pythagoras written block by block. The final conclusion is theorem 3.1. $\square$

Corollary 4.2 (Exact flatness gap)**.** *If $Z(A) \neq 0$ and $P_{\mathcal{B}}A \neq 0$, then*

$$\mathfrak{F}(P_{\mathcal{B}}A) - \mathfrak{F}(A) = \frac{|Z(A)|^2 (D(A) - D(P_{\mathcal{B}}A))}{D(P_{\mathcal{B}}A) D(A)}. \tag{16}$$

Thus the inflation is exactly controlled by the discarded within-block variance.

Proof. Subtract the two fractions with the common numerator $|Z(A)|^2$, and use (15). $\square$

4.1 Nested coarse-graining

Say that $\mathcal{B}_{\text{fine}}$ refines $\mathcal{B}_{\text{coarse}}$ if every fine block is contained in a coarse block. Then

$$\operatorname{Ran} P_{\text{coarse}} \subset \operatorname{Ran} P_{\text{fine}}.$$

Theorem 4.3 (Monotone certificate ladder)**.** *Let P_{fine} and P_{coarse} be the block projections of nested partitions, with the fine partition refining the coarse partition. Then*

$$D(P_{\text{coarse}}A) \le D(P_{\text{fine}}A) \le D(A), \tag{17}$$

and all three vectors have the same zero mode. On every nonzero branch,

$$\boxed{\mathfrak{F}(A) \le \mathfrak{F}(P_{\text{fine}}A) \le \mathfrak{F}(P_{\text{coarse}}A),} \tag{18}$$

with the analogous inequalities for ρ, all computed on the same frame Ω.

Proof. Because the ranges are nested,

$$P_{\text{coarse}}P_{\text{fine}} = P_{\text{coarse}}.$$

Apply norm contraction first to $P_{\text{fine}}A$, and then to A, to obtain (17). Both projections fix constants, hence preserve $Z(A)$. Dividing the common squared zero mode by the ordered energies proves (18). □

Remark 4.4 (Safe false negatives)*.* A coarse block certificate may fail because its denominator has lost too much within-block energy. Refining the partition can only improve the projected upper ratio. Conversely, if even a coarse level meets the desired upper bound, every finer level and the physical coefficient meet it. The ladder can therefore locate the coarsest provably sufficient resolution without risking a false positive.

4.2 Compression requires weights

The range of $P_{\mathcal{B}}$ has m degrees of freedom, but replacing it by an unweighted vector in $\mathbb{C}^m$ changes both the zero mode and the Hilbert norm. The correct compressed space is

$$H_{\mathcal{B}} = \mathbb{C}^m, \qquad \langle x, y\rangle_{\mathcal{B}} = \sum_{j=1}^{m} |B_j| x_j \overline{y_j}.$$

Proposition 4.5 (Weighted compressed isometry)**.** *The map*

$$C_{\mathcal{B}} : \operatorname{Ran} P_{\mathcal{B}} \to H_{\mathcal{B}}, \qquad C_{\mathcal{B}}(P_{\mathcal{B}}A) = (\overline{A}_1, \dots, \overline{A}_m),$$

is an isometry. In compressed coordinates the physical statistics are

$$Z_{\text{phys}} = \sum_{j=1}^{m} |B_j| \overline{A}_j, \tag{19}$$

$$D_{\text{phys}} = \sum_{j=1}^{m} |B_j| |\overline{A}_j|^2, \tag{20}$$

$$S_{\text{phys}} = \sum_{j=1}^{m} |B_j| = S. \tag{21}$$

These reproduce exactly the same-frame projected statistics.

Proof. Equations (20) and (21) give the norm identity, while (19) is (13). □

Example 4.6 (Unweighted compression manufactures cancellation)**.** *Let*

$$\Omega = \{1, 2, 3\}, \qquad B_1 = \{1\}, \quad B_2 = \{2, 3\},$$

and take the already block-constant vector

$$A = (1, -1, -1).$$

Then

$$Z_{\text{phys}} = 1 - 2 = -1, \qquad D_{\text{phys}} = 1 + 2 = 3, \qquad \mathfrak{F}_{\text{phys}} = \frac{1}{3}.$$

The unweighted compressed vector $(1, -1)$*, however, has apparent sum* 0 *and apparent flatness* 0*. No averaging error occurred: the false cancellation was created solely by deleting the multiplicities* $(1, 2)$*. Weighted compression restores the exact physical values.*

Remark 4.7 (Atomic normalization rule)*.* If literal atoms carry prescribed positive weights rather than unit weights, the same theorem holds in the weighted Hilbert space and block means must use those weights. Changing from physical atomic weights to unit block weights is a change of coefficient, not a harmless coordinate compression.

5 The missing-constant defect

We now drop the assumption $P\mathbf{1} = \mathbf{1}$. Retain

$$B = PA, \qquad R = (I - P)A, \qquad q = (I - P)\mathbf{1}, \qquad \kappa = \frac{\|q\|_2}{\sqrt{S}}.$$

Theorem 5.1 (Sharp zero-mode defect bound)**.** *For every orthogonal projection* P *and every* $A \in H$*,*

$$\boxed{Z(B) - Z(A) = \langle A, (P - I)\mathbf{1}\rangle = -\langle R, q\rangle,} \tag{22}$$

and hence

$$\boxed{|Z(B) - Z(A)| \leq \|R\|_2 \|q\|_2 = \sqrt{S}\,\kappa \|R\|_2.} \tag{23}$$

The coefficient is sharp. If $q \neq 0$*, equality holds for every prescribed residual norm when* R *is a scalar multiple of* q*.*

Proof. The identity is lemma 2.4; Cauchy–Schwarz gives the inequality. Given $t \geq 0$, choose

$$R = t\, e^{i\theta} \frac{q}{\|q\|_2}.$$

Then $R \in \operatorname{Ker} P$, $\|R\|_2 = t$, and

$$|\langle R, q\rangle| = t\|q\|_2.$$

Thus the constant cannot be improved from norm information alone. □

Remark 5.2 (The sharper residual norm)*.* The immediate estimate from the first pairing in (22) is

$$|Z(B) - Z(A)| \leq \|A\|_2 \|q\|_2.$$

Orthogonality removes the projected component and replaces $\|A\|_2$ by the smaller $\|R\|_2$. Both the constant defect and the discarded coefficient energy are therefore necessary in the sharp ledger.

Theorem 5.3 (Defective projection envelope)**.** *Assume* $B \neq 0$ *and put*

$$r = \frac{\|R\|_2}{\|B\|_2}.$$

Then

$$\boxed{\frac{(\rho(B) - \kappa r)_+}{\sqrt{1 + r^2}} \le \rho(A) \le \frac{\rho(B) + \kappa r}{\sqrt{1 + r^2}}.} \tag{24}$$

Equivalently, at the unnormalized zero-mode level,

$$\boxed{\frac{|Z(A) - Z(B)|}{\sqrt{S}\,\|B\|_2} \le \kappa r.} \tag{25}$$

The coefficient of κr in (25) is optimal.

Proof. The triangle and reverse triangle inequalities applied to

$$Z(A) = Z(B) + \langle R, q \rangle$$

give

$$(|Z(B)| - \sqrt{S}\,\kappa\|R\|_2)_+ \le |Z(A)| \le |Z(B)| + \sqrt{S}\,\kappa\|R\|_2.$$

Pythagoras gives

$$\|A\|_2 = \|B\|_2\sqrt{1 + r^2}.$$

Division proves (24). Equation (25) and its optimality follow from theorem 5.1. □

Corollary 5.4 (Product criterion)**.** *Let $\tau > 0$. The norm data guarantee*

$$|Z(A) - Z(B)| \le \tau\sqrt{S}\,\|B\|_2$$

whenever

$$\kappa(P)\frac{\|(I - P)A\|_2}{\|PA\|_2} \le \tau. \tag{26}$$

This product order is uniformly necessary: for $q \neq 0$, residuals parallel to q attain equality.

Proof. This is exactly (25), together with the equality construction in theorem 5.1. □

Example 5.5 (An unsafe projection can manufacture perfect cancellation)**.** *Let $S \ge 2$, let*

$$e = \frac{\mathbf{1}}{\sqrt{S}},$$

choose a unit vector $f \perp e$, and let P be the orthogonal projection onto $\mathbb{C}f$. Then

$$P\mathbf{1} = 0, \qquad \kappa(P) = 1.$$

For

$$A = f + re, \qquad r > 0,$$

one has

$$B = PA = f, \qquad \rho(B) = 0, \qquad \rho(A) = \frac{r}{\sqrt{1 + r^2}}.$$

Thus the projected vector reports perfect cancellation while the physical vector does not. For small r, the error is exactly first order:

$$\rho(A) = r + O(r^3).$$

Remark 5.6 (No monotonicity without the constant)*.* The unsafe projected energy still satisfies $D(B) \le D(A)$, but its numerator is a different zero mode. Energy contraction alone therefore gives no comparison between $\mathfrak{F}(A)$ and $\mathfrak{F}(B)$. Example 5.5 shows that the constant-preservation hypothesis in theorem 3.1 cannot be dropped.

6 Minimal rank-one constant completion

An unsafe orthogonal projection has a canonical repair. It adds exactly the direction in which the constant vector is missing.

Theorem 6.1 (Minimal constant completion)**.** *Let P be an orthogonal projection and let*

$$q = (I - P)\mathbf{1}.$$

If $q = 0$, put $P^\sharp = P$. If $q \neq 0$, define

$$P^\sharp x = Px + \frac{\langle x, q\rangle}{\|q\|_2^2} q. \tag{27}$$

Then:

(i) $P^\sharp$ is the orthogonal projection onto

$$\operatorname{Ran} P \oplus \mathbb{C}q = \operatorname{Ran} P + \mathbb{C}\mathbf{1};$$

(ii) $P^\sharp \mathbf{1} = \mathbf{1}$;

(iii) $\operatorname{Ran} P^\sharp$ is the smallest subspace containing $\operatorname{Ran} P$ and $\mathbf{1}$;

(iv) for every A,

$$D(P^\sharp A) = D(PA) + \frac{|\langle A, q\rangle|^2}{\|q\|_2^2}, \tag{28}$$

and

$$Z(P^\sharp A) = Z(A). \tag{29}$$

Proof. Because $q \in \operatorname{Ker} P = (\operatorname{Ran} P)^\perp$, the second term in (27) is the orthogonal projection onto $\mathbb{C}q$, perpendicular to P. Their sum is therefore the orthogonal projection onto $\operatorname{Ran} P \oplus \mathbb{C}q$.

Since $\mathbf{1} = P\mathbf{1} + q$,

$$\operatorname{Ran} P + \mathbb{C}\mathbf{1} = \operatorname{Ran} P \oplus \mathbb{C}q.$$

Also

$$\langle \mathbf{1}, q\rangle = \langle P\mathbf{1} + q, q\rangle = \|q\|_2^2,$$

so (27) gives

$$P^\sharp \mathbf{1} = P\mathbf{1} + q = \mathbf{1}.$$

Any subspace containing both $\operatorname{Ran} P$ and $\mathbf{1}$ contains $q = \mathbf{1} - P\mathbf{1}$, proving minimality. Orthogonality of the two summands in $P^\sharp A$ gives (28), and (29) follows from proposition 2.5. ☐

Corollary 6.2 (Repaired monotonicity)**.** *Let $A \neq 0$. If $P^\sharp A = 0$, then $Z(A) = 0$. Otherwise,*

$$\rho(A) \le \rho(P^\sharp A), \qquad \mathfrak{F}(A) \le \mathfrak{F}(P^\sharp A).$$

Proof. Apply theorem 3.1 to $P^\sharp$. ☐

Remark 6.3 (The repair is exact but not free)*.* The added coefficient is

$$\langle A, q\rangle = Z(A) - Z(PA).$$

Thus rank-one completion identifies the unique missing scalar, but does not estimate it. If that scalar is already available from an exact provenance ledger, the completion restores a safe same-frame projection. If it is the unknown arithmetic zero mode, writing $P^\sharp$ does not solve the arithmetic problem.

Proposition 6.4 (Uniqueness at minimal range)**.** *The operator $P^\sharp$ is the unique orthogonal projection Q such that*

$$\operatorname{Ran} Q = \operatorname{Ran} P + \mathbb{C}\mathbf{1}.$$

If $q \neq 0$, every constant-preserving orthogonal projection whose range contains $\operatorname{Ran} P$ has rank at least $\operatorname{rank} P + 1$, with equality precisely when its range equals $\operatorname{Ran} P^\sharp$.

Proof. An orthogonal projection is uniquely determined by its range. When $q \neq 0$, $\mathbf{1} \notin \operatorname{Ran} P$, so adjoining $\mathbf{1}$ raises the dimension by one. Every range containing both objects contains their span. ☐

7 The critical inverse-J interface

Let $\tau_j \to 0$ be a target angle scale. The TPC application will set $\tau_j = J_j^{-1}$.

Theorem 7.1 (Safe and defective transfer dichotomy)**.** *Let P_j be orthogonal projections on prescribed finite frames, let*

$$B_j = P_j A_j \neq 0, \qquad R_j = (I - P_j)A_j,$$

and put

$$\kappa_j = \frac{\|(I - P_j)\mathbf{1}\|_2}{\sqrt{S_j}}, \qquad r_j = \frac{\|R_j\|_2}{\|B_j\|_2}.$$

Suppose

$$\rho(B_j) \leq C\tau_j.$$

(i) If $\kappa_j = 0$, then

$$\rho(A_j) \leq C\tau_j$$

for arbitrary r_j.

(ii) If

$$\kappa_j r_j \leq K\tau_j,$$

then

$$\rho(A_j) \leq (C + K)\tau_j.$$

Proof. Part (i) is corollary 3.3. For part (ii), theorem 5.3 gives

$$\rho(A_j) \leq \frac{\rho(B_j) + \kappa_j r_j}{\sqrt{1 + r_j^2}} \leq (C + K)\tau_j.$$

□

Theorem 7.2 (Sharpness at the target scale)**.** *Uniform transfer of a zero-mode error target*

$$|Z(A) - Z(PA)| \leq C\tau\sqrt{S}\,\|PA\|_2$$

from only the two norm parameters $\kappa(P)$ and $r = \|(I - P)A\|_2/\|PA\|_2$ requires

$$\kappa(P)r = O(\tau)$$

in order. In particular, on any class where $\kappa(P) \geq \kappa_0 > 0$, the structure-free relative residual requirement returns as

$$r = O(\tau).$$

For $\tau = J^{-1}$, this is the TPC-89 critical order.

Proof. Sufficiency is corollary 5.4. For necessity, when $q = (I - P)\mathbf{1} \neq 0$, choose R parallel to q. Theorem 5.1 then gives the exact normalized defect

$$\frac{|Z(A) - Z(PA)|}{\sqrt{S}\,\|PA\|_2} = \kappa(P)r.$$

No smaller uniform order follows from the two norms. If $\kappa(P) \geq \kappa_0$, the stated residual order is necessary. □

Example 7.3 (Sharp angle failure beyond J^{-1})**.** *Use the projection from example 5.5, so $\kappa = 1$, and let*

$$B_J = f, \qquad A_J = f + r_J e.$$

Then

$$\rho(B_J) = 0, \qquad \rho(A_J) = \frac{r_J}{\sqrt{1 + r_J^2}}.$$

If Jr_J is unbounded, then $J\rho(A_J)$ is unbounded along a subsequence. Thus even a perfect projected zero mode need not transfer the critical angle when the missing-constant residual is larger than order J^{-1}.

Remark 7.4 (The product tradeoff)*.* The defective case does not always require $r = O(J^{-1})$. An approximately constant-preserving projection with $\kappa = o(1)$ can tolerate a larger residual, provided

$$\kappa r = O(J^{-1}).$$

The two factors have different meanings: κ is an operator defect known before seeing A, while r measures the discarded energy of the particular coefficient. They should be recorded separately before their product is compared with the target.

7.1 Unnormalized critical scale

The same condition appears in the physical zero-mode normalization. Assume along a critical family that

$$S_X \le X^{o(1)} Q_X, \qquad D(B_X) \le X^{o(1)} Q_X^3 J_X^2, \tag{30}$$

where $B_X = P_X A_X \ne 0$.

Proposition 7.5 (Defect at the Q^2 scale)**.** *Under (30), put*

$$\kappa_X = \frac{\|(I - P_X)\mathbf{1}\|_2}{\sqrt{S_X}}, \qquad r_X = \frac{\|(I - P_X)A_X\|_2}{\|P_X A_X\|_2}.$$

Then

$$|Z(A_X) - Z(B_X)| \le X^{o(1)} \kappa_X r_X Q_X^2 J_X. \tag{31}$$

Hence a sufficient condition for

$$|Z(A_X) - Z(B_X)| \le X^{-\eta z + o(1)} Q_X^2$$

is

$$\boxed{\kappa_X r_X \le \frac{X^{-\eta z + o(1)}}{J_X}.} \tag{32}$$

For a constant-preserving projection, the defect is identically zero and this approximation condition disappears.

Proof. By theorem 5.1,

$$|Z(A_X) - Z(B_X)| \le \sqrt{S_X}\, \kappa_X r_X \sqrt{D(B_X)}.$$

Insert (30) to obtain (31); the final statement follows immediately. □

Remark 7.6 (Zero-mode and energy goals remain distinct)*.* Exact zero-mode preservation does not by itself prove that the projected energy is large. Conversely, a large projected energy does not estimate its zero mode. Proposition 3.6 shows that proved projected bounds transfer in the correct directions, but both inputs still require their own arguments.

8 Literal fixed-shift crosswalk

Fix one prescribed integer $h_0 \neq 0$. At the high-β endpoint, retain the critical scales

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}. \tag{33}$$

Let A_X be the already aggregated fixed-h_0 determinant coefficient from TPC-32, on its actual post-bin support

$$\Omega_X = \operatorname{supp} A_X, \qquad S_X = \#\Omega_X \ll X^{o(1)} Q_X$$

[2]. Put

$$Z_X = Z(A_X), \qquad D_X = D(A_X), \qquad \mathfrak{F}_X = \frac{|Z_X|^2}{D_X}$$

on the nonzero branch.

Definition 8.1 (Literal orthogonal surrogate)**.** A literal orthogonal surrogate is a vector

$$B_X = P_X A_X \in \mathbb{C}^{\Omega_X},$$

where:

(i) P_X is an orthogonal projection for the original global atomic inner product;

(ii) every output coordinate is mapped to the same physical determinant bin as its input coordinate;

(iii) no native row, channel, mask, content, resonance, short-fiber, or polarization sector is deleted before the map; and

(iv) if coordinates are compressed, their exact physical multiplicities are retained as in proposition 4.5.

It is *zero-mode safe* if $P_X \mathbf{1} = \mathbf{1}$.

Theorem 8.2 (Literal safe-projection certificate)**.** *Let $B_X = P_X A_X$ be a zero-mode-safe literal orthogonal surrogate. Then*

$$Z(B_X) = Z_X, \qquad D(B_X) \leq D_X. \tag{34}$$

If $B_X = 0$, then $Z_X = 0$ and $\mathfrak{F}_X = 0$ whenever $A_X \neq 0$. If $B_X \neq 0$ and

$$\rho(B_X) \leq \frac{X^{o(1)}}{J_X}, \tag{35}$$

then

$$\boxed{\rho(A_X) \leq \frac{X^{o(1)}}{J_X}, \qquad \mathfrak{F}_X \leq X^{1/400+o(1)}.} \tag{36}$$

No relative approximation hypothesis between A_X and B_X is needed.

Proof. Equation (34) and the zero branch follow from theorem 3.1. On the nonzero branch,

$$\rho(A_X) \leq \rho(B_X).$$

Therefore (35) gives the first conclusion in (36). Using $\mathfrak{F}_X = S_X \rho(A_X)^2$,

$$\mathfrak{F}_X \leq X^{o(1)} \frac{Q_X}{J_X^2} = X^{(267-266)/400+o(1)} = X^{1/400+o(1)}.$$

□

Corollary 8.3 (Projected energy certificate)**.** *Under the hypotheses of theorem 8.2, a proved lower bound*

$$D(B_X) \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2$$

implies the same lower bound for D_X. If a projected zero-mode estimate

$$|Z(B_X)| \leq X^{-\eta_Z+o(1)} Q_X^2$$

is also proved, then the physical coefficient inherits it exactly.

Proof. Use (34). □

Remark 8.4 (What changed relative to TPC-89)*.* For an arbitrary same-frame surrogate, TPC-89 requires

$$\frac{\|A_X - B_X\|_2}{\|B_X\|_2} \leq \frac{X^{o(1)}}{J_X}$$

to transfer (35) in the worst case [4]. The present theorem removes that requirement only for the narrower class

$$B_X = P_X A_X, \qquad P_X = P_X^* = P_X^2, \qquad P_X \mathbf{1} = \mathbf{1}.$$

It is a structural bypass, not an improvement of the general perturbation constant.

8.1 Interfaces with the low-window export

TPC-93 exports the literal low-frequency zero-mode packet into decorated affine fibers, retaining content factors, resonant and short sectors, both polarizations, and the fixed determinant [6]. This creates possible block labels, but does not automatically define a projection on $\mathbb{C}^{\Omega_X}$.

Proposition 8.5 (Promotion checklist)**.** *A grouping derived from the TPC-93 affine ledger may invoke theorem 8.2 only after proving:*

(1) a lossless map from every decorated source atom back to its physical determinant bin;

(2) that the proposed operation is an orthogonal projection for the single global atomic normalization;

(3) that it fixes the physical constant vector, or else that the defect product in (32) is met;

(4) that compressed block coordinates carry their exact multiplicities;

(5) that all native keys and every short, constant, resonant, and generic sector are returned; and

(6) an actual estimate for the projected coefficient at the required growing fixed-h_0 scale.

Proof. Items 1–4 place the operator in the Hilbert space of definition 8.1; item 5 ensures it is the complete coefficient rather than a deleted-sector model. Only then does item 6 supply the hypothesis (35) or the separate projected zero-mode/energy bounds. Without any item, the relevant theorem concerns a different coefficient or remains conditional. □

Remark 8.6 (Frequency averaging is not bin projection)*.* Averaging Fourier frequencies, affine origins, shifts, or decorated fibers is not, by itself, a projection on the physical determinant bins. A literal inverse map must show that the operation equals some P_X on $\mathbb{C}^{\Omega_X}$. Otherwise the projected monotonicity theorem does not apply.

8.2 Independent endpoint ledgers

If future work proves both exponents in corollary 8.3, the determinant compatibility remains

$$\boxed{\lambda_D \leq 2\eta_Z.} \tag{37}$$

Separately, all physical fixed-power costs not already recorded in $\lambda_D - 2\eta_Z$ must satisfy

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}.} \tag{38}$$

Projection monotonicity itself incurs no exponent loss, but neither ledger may be inferred from that fact. Equality in (37) has no determinant reserve; equality in (38) stops the physical route [5].

9 Executable finite certificate

The script

`experiments/check_projection_identities.py`

is a finite regression certificate for the algebraic statements in this paper. It uses exact rational arithmetic to check:

(i) every set partition of frames of size at most five and every coefficient in $\{-2,-1,0,1,2\}^{\Omega}$, excluding only the all-zero coefficient, for zero-mode preservation, Pythagoras, within-block variance, and flatness monotonicity;

(ii) every coordinate projection on frames of size at most five for the exact constant-defect identity, the sharp Cauchy bound, and rank-one constant completion; and

(iii) the weighted versus unweighted three-point counterexample.

It also generates seeded random complex orthogonal projections. Safe projectors are built by including the normalized constant vector in an orthonormal basis; defective projectors are repaired by the rank-one formula in (27). The random checks cover the full identities, norm monotonicity, defect bound, and repaired zero-mode preservation.

The archived run completed

$$172{,}480$$

exact block cases,

$$111{,}110$$

exact coordinate-projection cases, including

$$107{,}205$$

nontrivial sharp-equality residuals, and 1,000 seeded random cases in each of the safe and defective classes. Every check passed.

Remark 9.1 (Role of the computation). The script is not evidence for any asymptotic arithmetic claim. It guards orientation, conjugation, block-weight, zero-branch, and normalization conventions in exact finite models. The proofs in the preceding sections are the mathematical justification.

10 Claim boundary and next gate

Theorem 10.1 (Audited conclusion)**.** *Let P be an orthogonal projection on a prescribed finite actual-support frame, let $B = PA$, and let $R = (I - P)A$.*

(i) If $P\mathbf{1} = \mathbf{1}$, then

$$Z(B) = Z(A), \qquad D(A) = D(B) + D(R),$$

and every nonzero projected branch satisfies

$$\rho(A) \le \rho(B), \qquad \mathfrak{F}(A) \le \mathfrak{F}(B).$$

The zero projected branch forces $Z(A) = 0$.

(ii) For arbitrary P,

$$Z(B) - Z(A) = -\langle R, (I - P)\mathbf{1}\rangle$$

and

$$|Z(B) - Z(A)| \le \sqrt{S}\,\kappa(P)\|R\|_2$$

with optimal constant.

(iii) If $B \neq 0$, the critical structure-free condition is

$$\kappa(P)\frac{\|R\|_2}{\|B\|_2} = O(J^{-1}).$$

It is vacuous for an exactly constant-preserving projection and returns the TPC-89 J^{-1} residual order when $\kappa(P) \asymp 1$.

(iv) Every unsafe P has the unique minimal constant completion $P^\sharp$ from (27).

For the literal fixed-h_0 TPC coefficient, a zero-mode-safe orthogonal surrogate meeting the projected critical angle bound certifies

$$\mathfrak{F}_X \le X^{1/400+o(1)}$$

without a relative approximation hypothesis.

Proof. Combine theorems 3.1, 5.1, 6.1, 7.2 and 8.2. □

What is proved at L0. The projection ledger, exact monotonicity, equality cases, partition variance law, nested hierarchy, weighted compression theorem, unweighted counterexample, sharp defect bound, product threshold, and rank-one completion are unconditional finite-dimensional theorems.

What is proved at L1. Once an explicitly constructed operator is verified on the literal post-bin frame, the paper gives a proof-carrying certificate that transfers projected zero-mode and energy bounds to the physical coefficient in the correct directions. The checklist in proposition 8.5 is part of this interface.

What is not proved at L2. No constant-preserving projection with a small projected flatness is constructed for the growing arithmetic family. No projected determinant-energy lower bound, low-window cancellation estimate, affine Möbius-correlation theorem, full-block return, or fixed-shift asymptotic is proved.

No free averaging. Block averaging cannot manufacture a smaller physical flatness when it is performed as a same-frame constant-preserving orthogonal projection. It can manufacture apparent cancellation if block multiplicities are deleted, if the operator misses the constant mode, or if an ensemble/frequency average is substituted for the physical bins. Those operations are outside the safe theorem.

No contradiction with the accuracy barrier. The TPC-89 barrier concerns arbitrary error directions. Constant-preserving orthogonal projection removes precisely the zero-mode component of the error. In the defective case, the sharp product κr restores the same critical order. These are complementary statements.

Arithmetic boundary. Nothing here specializes the prescribed nonzero h_0 to 2, proves a growing fixed-shift Möbius or Liouville correlation, crosses the sieve parity barrier, proves a Hardy–Littlewood asymptotic, gives a prime-pair lower bound, or proves the twin-prime conjecture.

Next gate. The most useful continuation is a literal projection census:

(1) define candidate blocks from the native TPC-32/TPC-93 keys;

(2) verify their exact inverse map, global weights, and $P_X\mathbf{1} = \mathbf{1}$;

(3) compute the projected zero mode and energy without deleting any sector;

(4) prove or disprove

$$\rho(P_X A_X) \le \frac{X^{o(1)}}{J_X}$$

for one growing family; and

(5) if constants are not preserved, test the sharp fallback

$$\kappa_X r_X \le \frac{X^{o(1)}}{J_X}.$$

A positive estimate would promote the projection route. A lower bound showing every admissible coarse projection has excessive flatness would be a precise negative certificate and would redirect the program toward the low-window/Tauberian or full-block routes.

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